

Adaptive Risk Allocation in Crypto Markets: Evaluating Volatility-Scaled Portfolios

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Abstract

This study applies volatility scaling, an established timing strategy that dynamically adjusts portfolio risk exposure, to the cryptocurrency market. We scale the market portfolio and seven cryptocurrency strategies by their recent volatility to improve risk-adjusted returns. Volatility scaling increases Sharpe ratios and generates abnormal returns in certain strategies, particularly momentum-based strategies and the overall market portfolio. The strategy remains effective even under tight leverage constraints and its advantages become more evident as we extend the investment horizon. Our results are robust after accounting for common risk factors and limits to arbitrage. We confirm that similar to the equity market, volatility scaling does not systematically benefit all strategies, and the strongest effects occur in momentum portfolios, suggesting a robust and potentially universal pattern.

Keywords: Cryptocurrency, Volatility, Portfolio, Momentum, Market Timing, Risk-return trade-off, Asset pricing

1 Introduction

The relationship between risk and return is fundamental to asset pricing theory, but empirical findings sometimes indicate more complex dynamics than traditional models. For instance, in both cryptocurrency and equity markets volatility has strong persistence, but a weak correlation with future returns. To highlight this pattern in the cryptocurrency market, we plot next week’s volatilities and mean excess returns versus terciles of the current week’s volatilities in Figure 1. The left-hand panel shows a strong relationship between current and next week’s volatilities, but the right-hand panel shows no clear pattern in returns across different volatility terciles. This implies that, despite predictably higher risk, expected returns do not offer proportionally higher rewards. Therefore, risk-adjusted returns could potentially increase if we take more risk when recent volatility is low and less risk when recent volatility is high.

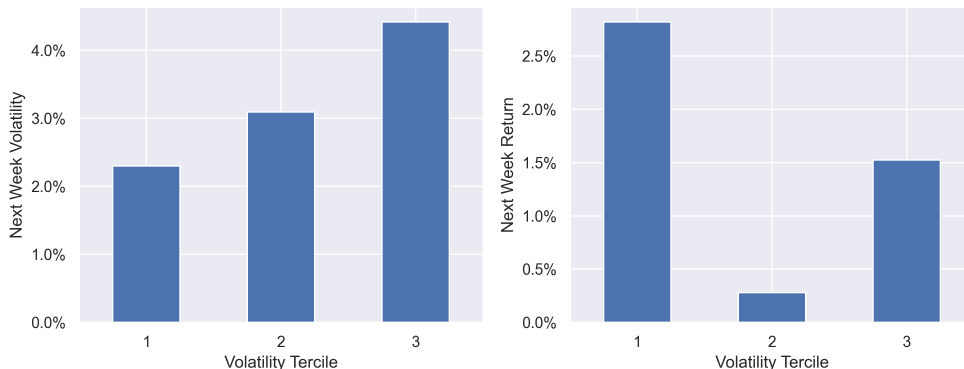


Figure 1: Weekly Market Portfolio Performance by Prior-week’s Volatility Terciles: Weeks are grouped into terciles based on the previous week’s realized volatility of the market portfolio, showing the average standard deviation and return for each tercile in the following week. (cf. Harvey et al. (2018) for other asset classes.)

Volatility scaling is a systematic approach to implementing such dynamic risk adjustments, by scaling portfolio leverage inversely to its recent volatility. This only affects the total capital allocation over time, while keeping the portfolio composition unchanged.

Several studies have shown the potential of volatility scaling to enhance performance, though its benefits vary by strategy and market. Barroso and Santa-Clara (2015) and Daniel

and Moskowitz (2016) show that scaled momentum strategies nearly double the Sharpe ratio of the original momentum strategy. Moreira and Muir (2017) find that scaled strategies earn systematically positive alphas relative to their unscaled counterparts across a wide range of factor portfolios. However, Cederburg et al. (2020) emphasize that in fact, spanning regression alphas point to optimal combinations of scaled and unscaled portfolios, rather than the direct superiority of scaled portfolios. Therefore, they rely on direct Sharpe ratio comparison and find that scaled portfolios do not systematically outperform their unscaled versions, and statistically significant improvements are mostly present in momentum-related strategies. Harvey et al. (2018) extend the study to other asset classes and report benefits in equity and credit portfolios but find negligible effects in fixed income, currencies, and commodities markets.

Despite cryptocurrency’s extreme volatility and growing institutional interest, volatility scaling isn’t explored in this market. We address this gap by examining volatility scaling across eight cryptocurrency trading strategies: the market portfolio and seven well-documented characteristic-based long-short portfolios.

We contribute to the literature in three ways: First, we extend the application of volatility scaling to a new and increasingly important asset class and show that it substantially enhances specific strategies. Second, we show that even the total market portfolio can benefit from volatility-scaling, as the scaled market portfolio generates positive alpha in the spanning regression on its unscaled counterpart. Finally, we provide strong support for the effectiveness of volatility scaling in momentum strategies, reinforcing similar findings from the equity market and demonstrating their robustness in a different market.

The remainder of this paper is organized as follows: Section 2 describes our data and methodology, Section 3 presents our empirical findings, Section 4 describes our robustness checks, and Section 5 concludes.

2 Data and Methodology

2.1 Data

We collected historical daily data for all cryptocurrencies available in CoinMarketCap¹, excluding coins with no market capitalization data. Following Long et al. (2020), we further excluded observations with non-positive market values. The final sample consists of close price, market capitalization, and trading volume data for 3,061 coins from 2016 to 2022.² For the risk-free rate, we use U.S. T-bill returns as reported in French (2023).

2.2 Portfolio Construction

Our base strategies consist of 8 cryptocurrency strategies. To select these strategies, we examined 13 long-short strategies introduced in prior studies and chose only those with statistically significant positive returns over the full seven-year period. Due to the high volatility of cryptocurrency markets, we employ weekly holding periods instead of monthly, in line with common cryptocurrency literature practices to enhance estimation accuracy.

The market index return is the capitalization-weighted average of all cryptocurrency returns. We subtract the risk-free rate to find the excess return of the cryptocurrency market portfolio (CMKT). We also construct seven long-short portfolios as identified by Liu et al. (2022). They are based on specific cryptocurrency characteristics, defined in table [1](#).

¹Accessed: October 2, 2023

²Data prior to 2016 is used when necessary to calculate specific variables.

Table 1: Characteristic Definitions

Category	Factor	Definition
Size	MCAP	Log last day market capitalization in the portfolio formation week
	PRC	Log last day price in the portfolio formation week
Momentum	MAXDPRC	Maximum price of the portfolio formation week
	r_1	One-week momentum
	r_2	Two-week momentum
	r_3	Three-week momentum
Volatility	STDPRCVOL	Log standard deviation of dollar volume in the portfolio formation week

Notes: Definitions of characteristics used for constructing long-short portfolios grouped into Size, Momentum, and Volatility categories.

To construct each portfolio, we use univariate portfolio analysis. At the end of the portfolio formation weeks, coins with market capitalizations below \$1 million or zero trading volumes are excluded. Then, we sort cryptocurrencies into quintiles based on the value of the corresponding characteristic and form a value-weighted portfolio for each quintile. The following week’s returns for each quintile are computed, and excess returns are obtained by subtracting the risk-free rate. Each long-short portfolio is created by taking a long position in the highest (fifth) quintile and a short position in the lowest (first) quintile³. The strategy’s excess return is equal to the difference between these two quintiles’ excess returns.

2.3 Volatility-Scaling

We implement volatility scaling by adjusting each portfolio’s exposure inversely to its realized standard deviation of returns from the prior week. For a given portfolio with excess return R_t in week t , we construct its scaled version (R_t^s) as follows:

$$R_t^\sigma = \frac{c^*}{\sigma_{t-1}} R_t \quad (1)$$

³For size and volatility strategies, where the return relationship is negative, we reverse this approach: taking a long position in the first quintile and a short position in the fifth.

where σ_{t-1} is the realized standard deviation⁴ of daily returns in the previous week, and c^* is a scaling constant determined such that the scaled portfolio maintains the same unconditional volatility as the original portfolio over the entire sample period.⁵

In this equation, the dynamic investment position in each portfolio, $\frac{c^*}{\sigma_{t-1}}$, measures the leverage applied in constructing the scaled portfolio in week t . Importantly, volatility scaling does not change the relative weights of assets in the portfolio; rather, it adjusts the capital allocated over time to stabilize the portfolio’s risk exposure.

3 Empirical Results

3.1 Base Strategy Performance

We begin with the baseline performance of strategies. Table 2 presents excess returns for quintile portfolios and the associated long-short implementations. All strategies generate statistically significant returns at the 10% level, and the market portfolio (CMKT) delivers an excess return of 1.89% (t-stat = 2.95), which indicates a substantial risk premium in the cryptocurrency market.

⁴While either variance or standard deviation could serve as a volatility proxy, we follow Barroso and Santa Clara (2015), Harvey et al. (2018), and Wang and Yan (2021) and use standard deviation primarily because it leads to less extreme investment weights and hence lower turnover and trading cost. Our results are qualitatively similar if variance is used, though statistical significance is somewhat reduced.

⁵The scaling parameter is not known to an investor in real time, but Sharpe ratios are invariant to the choice of c^* .

Table 2: Performance of Quintile and Long-Short Portfolios

Strategy	CMKT	MCAP	PRC	MAXDPRC	r_1	r_2	r_3	STDPRCVOL
1		0.064	0.111	0.111	0.024	0.010	0.016	0.080
2		0.027	0.031	0.029	0.008	0.013	0.008	0.031
3		0.025	0.017	0.017	0.020	0.014	0.020	0.038
4		0.035	0.017	0.017	0.038	0.032	0.031	0.034
5		0.021	0.017	0.017	0.037	0.039	0.034	0.018
Long-Short	0.019	0.043	0.093	0.094	0.013	0.029	0.018	0.062
	(2.95)	(2.83)	(2.21)	(2.25)	(2.82)	(3.62)	(2.63)	(1.80)

Notes: Average weekly excess returns for quintile-sorted portfolios and long-short implementations, with corresponding (t-statistics). For characteristics with positive return relationships (Momentum category), we long quintile 5 and short quintile 1; for negative relationships, we long quintile 1 and short quintile 5.

3.2 Impact of Volatility Scaling

We find that volatility scaling significantly improves specific strategies, including market capitalization (MCAP) and three momentum-based strategies (one-week, two-week, and three-week momentum), as detailed in Panel A of Table 3. Its impact on other strategies is not significant at the 10% level. Momentum strategies display the most pronounced improvements, with Sharpe ratios increasing about 0.43–0.44 units (p-values < 0.05).

Implementing volatility scaling requires access to leverage. To assess its effectiveness under leverage constraints, we evaluated strategies with leverage caps of 3 $\times$ and 1 $\times$ (no leverage). In high-volatility periods, both scenarios adjust leverage similarly to the main implementation. However, when the main implementation assigns leverage higher than 3 $\times$ or 1 $\times$, the leverage is capped at 3 $\times$ and 1 $\times$, respectively.⁶ Results indicate that even under stringent leverage caps, volatility scaling remains beneficial. With a maximum of 3 $\times$ leverage, all four successful strategies achieve Sharpe ratio improvements. Under a 1 $\times$ cap, one-week and three-week momentum strategies, as well as MCAP, still exhibit Sharpe ratio gains. This suggests that the benefits primarily derive from risk reduction during volatile periods rather than leverage.

⁶The 1 $\times$ (no leverage) scenario implies reducing exposure during high-volatility periods, without changing exposure in low volatility periods.

Further insights are provided by our spanning regression analysis (Panel B, Table 3), where we regress excess returns of volatility-scaled portfolios on their unscaled counterparts:

$$R_t^\sigma = \alpha + \beta R_t + \varepsilon_t. \quad (2)$$

Here, a non-zero alpha implies that the optimal ex-post combination of scaled and unscaled portfolios, with a positive weight on the scaled version, expands the efficient frontier beyond what can be achieved by unscaled portfolios alone (Cederburg et al., 2020). Generally, Panel B results are similar to Panel A. Seven of the eight portfolios display positive alphas, and five reach statistical significance. These include the four previously successful strategies and the market portfolio, which demonstrates an alpha of 0.90% (t-stat = 2.23) compared to its original version, highlighting the broader potential of volatility scaling.

Constructing a volatility-scaled portfolio is straightforward in real time, but realizing the full benefit of these alphas can be challenging in real time, as it requires combining scaled and unscaled portfolios with ex-post optimal weights, which are inherently complex to determine (Cederburg et al., 2020). This is why we primarily performed a direct Sharpe ratio comparison. However, as proposed by Wang and Yan (2021), we find that a simple 50/50 fixed-weight allocation to scaled and unscaled portfolios effectively captures a substantial portion of the benefit, resulting in a 0.21 increase in the Sharpe ratio for the market portfolio (p-value = 0.046).

In panel C, we control for the cryptocurrency three-factor model of Liu et al. (2022) using their three factors as control variables. The results remain consistent with panels A and B. All successful strategies still generate significant risk-adjusted returns. This indicates that volatility scaling’s benefits stem from a distinct mechanism rather than exposure to known risk factors.

Table 3: Impact of Volatility Scaling

	CMKT	MCAP	PRC	MAXDPRC	r_1	r_2	r_3	STDPRCVOL
Panel A: Sharpe Ratios								
SR_{original}	1.112	1.069	0.835	0.850	1.066	1.366	0.994	0.677
SR_{scaled}	1.399	1.419	0.656	0.769	1.497	1.801	1.437	0.687
$SR_{\text{scaled}} - SR_{\text{original}}$	0.287	0.350	-0.180	-0.081	0.431	0.435	0.443	0.009
	[0.123]	[0.077]	[0.797]	[0.645]	[0.021]	[0.023]	[0.024]	[0.490]
$SR_{\text{scaled}}(\text{lev}_{\text{max}} = 3\times)$	1.384	1.403	0.716	0.796	1.478	1.743	1.393	0.694
$SR_{\text{scaled}}(\text{lev}_{\text{max}} = 3\times) - SR_{\text{original}}$	0.272	0.333	-0.120	-0.054	0.412	0.377	0.399	0.017
	[0.124]	[0.070]	[0.718]	[0.603]	[0.020]	[0.038]	[0.030]	[0.482]
$SR_{\text{scaled}}(\text{lev}_{\text{max}} = 1\times)$	1.086	1.238	0.714	0.751	1.371	1.541	1.283	0.788
$SR_{\text{scaled}}(\text{lev}_{\text{max}} = 1\times) - SR_{\text{original}}$	-0.026	0.169	-0.121	-0.099	0.306	0.174	0.289	0.111
	[0.567]	[0.029]	[0.746]	[0.709]	[0.009]	[0.125]	[0.032]	[0.369]
Panel B: Spanning Tests								
Alpha (α)	0.009	0.023	-0.005	0.006	0.017	0.017	0.015	0.033
	(2.23)	(2.45)	(-0.21)	(0.28)	(2.92)	(3.18)	(2.84)	(1.09)
Panel C: Three-Factor Control								
Alpha (α)	0.009	0.022	-0.029	-0.017	0.019	0.020	0.018	-0.014
	(2.29)	(2.46)	(-1.27)	(-0.74)	(3.28)	(3.66)	(3.41)	(-0.49)

Notes: Panel A compares annualized Sharpe ratios for scaled and unscaled portfolios, including results with leverage caps. [p-values] are from the Jobson and Korkie (1981) test of the null that $SR_{\text{scaled}} - SR_{\text{original}} > 0$, adjusted by Memmel (2003). Panel B presents spanning test alphas (t-statistics), and Panel C shows alphas (t-statistics) after controlling for the cryptocurrency three-factor model.

Next, we examine the stability of the observed benefits across different time horizons. Figure 2 shows the outperformance percentage of scaled strategies across rolling windows ranging from 25 to 250 weeks⁷. For each window length, this is the proportion of all periods of that length in which the scaled portfolio has a higher Sharpe ratio than its base portfolio. Across all timeframes, all four strategies (MCAP and the three momentum-based) show superior performance, which improves over time, and ultimately reaches 100% outperformance. Momentum strategies benefit from volatility scaling more quickly and consistently, and the MCAP strategy improves more gradually. Based on these patterns, volatility scaling has structural rather than transient advantages, which become more pronounced over the long term. However, the performance advantage of the scaled market portfolio tends to be more irregular across time horizons, indicating variability in the scaling efficacy of the market portfolio.

⁷Our dataset spans 370 weeks, allowing for the analysis of at least 120 distinct rolling windows of 250 weeks.

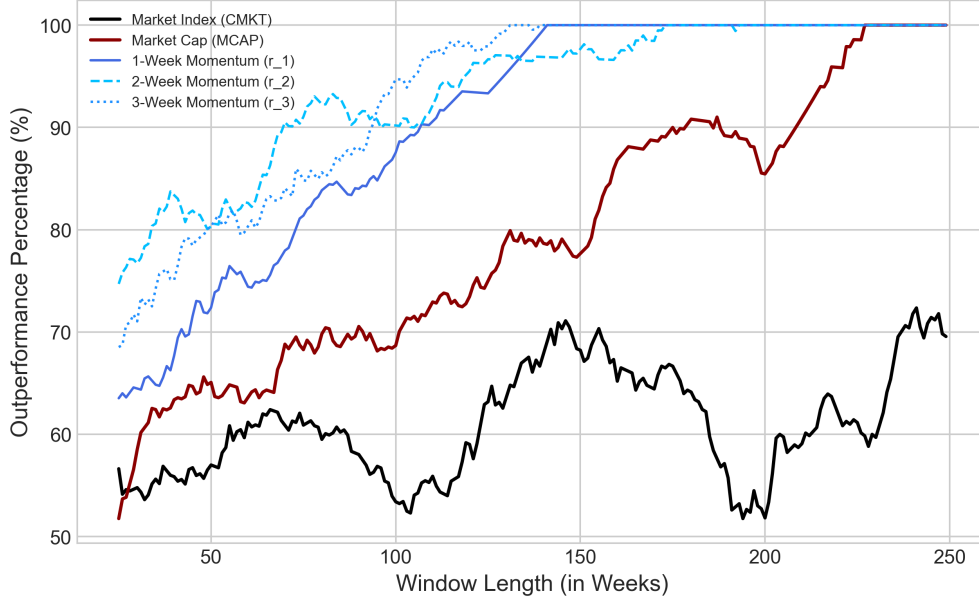


Figure 2: Rolling Performance of Volatility-Scaled Strategies: Percentage of periods where scaled portfolios outperform their unscaled counterparts across different investment horizons (25-250 weeks) for CMKT and successful strategies (MCAP, r_1 , r_2 , r_3).

4 Robustness Checks

We employed a simple scaling method to exploit the mispricing shown in figure 1, which resulted in improved risk-adjusted returns. However, in the real-world, market frictions may impose limits on arbitrage and prevent market participants from fully exploiting such mispricings. In this section, we explore how illiquidity, short-selling impediments, and transaction costs affect volatility scaling to ensure our findings are valid and practical.

Liquidity constraints directly limit arbitrage opportunities and may influence our results. Our scaling method may appear beneficial in part due to illiquid coins. In these coins, mispricings are more difficult to correct because of low trading volumes, difficulty in executing large trades without impacting prices, and wider bid-ask spreads. To address these concerns, Table 4 presents Sharpe ratio improvements across Amihud’s (2002) illiquidity quintiles. In general, scaling benefits for momentum strategies are highly concentrated in more liquid market segments (quintiles 1–3), thus practical limitations are not expected to prevent these

strategies from capturing scaling benefits. In contrast, the MCAP strategy shows stronger improvements in less liquid segments (quintiles 3 and 5), suggesting that illiquidity may both act as a barrier to strategy implementation and as a driver of higher risk-adjusted returns. However, for the CMKT portfolio, no clear pattern exists across the liquidity quintiles. This indicates that liquidity does not systematically affect scaling benefits for the market portfolio.

Table 4: Scaling Benefits Across Illiquidity Quintiles

	CMKT	MCAP	PRC	MAXDPRC	r_1	r_2	r_3	STDPRCVOL
1	0.112 [0.29]	-0.166 [0.81]	-0.472 [0.94]	-0.434 [0.92]	0.420 [0.04]	0.130 [0.30]	0.493 [0.02]	-0.290 [0.86]
2	-0.105 [0.72]	-0.463 [0.97]	0.320 [0.06]	0.290 [0.07]	0.749 [0.001]	0.359 [0.08]	0.220 [0.20]	0.278 [0.09]
3	-0.063 [0.66]	0.917 [0.004]	-0.004 [0.51]	-0.007 [0.51]	0.349 [0.03]	0.510 [0.006]	0.284 [0.06]	0.189 [0.18]
4	-0.568 [0.99]	0.220 [0.18]	0.052 [0.42]	0.043 [0.44]	-0.159 [0.80]	-0.460 [0.98]	0.045 [0.41]	0.224 [0.23]
5	-0.133 [0.68]	1.099 [0.001]	0.404 [0.000]	0.252 [0.000]	0.044 [0.38]	-0.425 [1.00]	-0.038 [0.59]	0.955 [0.007]

Notes: Annualized Sharpe ratio differences between scaled and unscaled portfolios across Amihud illiquidity quintiles, with [p-values] from the Jobson and Korkie (1981) test of the null that $SR_{\text{scaled}} - SR_{\text{original}} > 0$, adjusted by Memmel (2003).

Short-selling impediments are another potential limitation to arbitrage when implementing long-short portfolios. Although short-selling was not universally available for all cryptocurrencies in our sample, we do not expect it to seriously undermine our conclusions. First, the base strategies are capitalization-weighted. They assign higher weights to large-cap coins, which are generally more accessible for short-selling. Similarly, the short leg of the MCAP strategy consists of large-cap coins. Moreover, improvements in momentum strategies tend to come from the most liquid market segments, where short-selling impediments are minimal. Therefore, while it may pose some implementation challenges, this will not meaningfully diminish the efficacy of volatility scaling.

However, transaction costs may still erode scaling benefits, due to the increased turnover⁸

⁸Turnovers are calculated in accordance with Barroso and Detzel (2021).

caused by dynamic scaling. To assess this, we analyze Sharpe ratios using net-of-cost returns, with a transaction cost of 25 basis points, following Brauneis and Mestel (2019). Table 5 presents these adjusted results, showing that while our main findings are slightly weaker, they remain robust. In unreported results, we also found that the scaled CMKT portfolio continues to generate significant positive alpha after costs. Moreover, momentum strategies improvements can survive higher transaction costs, but this could materially impact the net-of-cost performance of the scaled MCAP strategy.

Table 5: Sharpe Ratios Adjusted for Transaction Costs

	CMKT	MCAP	PRC	MAXDPRC	r_1	r_2	r_3	STDPRCVOL
SR_{original}	1.112	1.050	0.832	0.847	0.916	1.244	0.881	0.669
SR_{scaled}	1.338	1.313	0.610	0.724	1.280	1.620	1.266	0.602
$SR_{\text{scaled}} - SR_{\text{original}}$	0.226 [0.149]	0.262 [0.113]	-0.222 [0.877]	-0.123 [0.739]	0.364 [0.025]	0.376 [0.024]	0.385 [0.025]	-0.068 [0.578]
$SR_{\text{scaled}}(\text{lev}_{\text{max}} = 3\times)$	1.340	1.329	0.697	0.777	1.263	1.569	1.228	0.660
$SR_{\text{scaled}}(\text{lev}_{\text{max}} = 3\times) - SR_{\text{original}}$	0.229 [0.135]	0.279 [0.080]	-0.135 [0.769]	-0.069 [0.648]	0.347 [0.025]	0.325 [0.041]	0.346 [0.032]	-0.009 [0.511]
$SR_{\text{scaled}}(\text{lev}_{\text{max}} = 1\times)$	1.065	1.205	0.709	0.745	1.205	1.404	1.151	0.770
$SR_{\text{scaled}}(\text{lev}_{\text{max}} = 3\times) - SR_{\text{original}}$	-0.047 [0.635]	0.155 [0.025]	-0.123 [0.778]	-0.101 [0.737]	0.290 [0.006]	0.160 [0.116]	0.270 [0.025]	0.101 [0.366]

Notes: Annualized Sharpe ratios for scaled and unscaled strategies adjusted for a 25 basis-point transaction cost, including results with leverage caps. [p-values] are from the Jobson and Korkie (1981) test of the null that $SR_{\text{scaled}} - SR_{\text{original}} > 0$, adjusted by Memmel (2003).

5 Conclusion

The overall portfolio of the cryptocurrency market benefits from volatility scaling, as its scaled version produces abnormal returns over its original counterpart, which illustrates the wide applicability of volatility scaling. The benefits of volatility scaling vary among other portfolios, and momentum strategies experience the greatest gains in risk-adjusted returns. This reinforces previous equity studies and suggests that the relation between momentum strategies and volatility is more likely to be a deeper, cross-market phenomenon rather than a market-specific one. Furthermore, we show that scaling advantages persist even after accounting for liquidity constraints, transaction costs, and other practical limitations. This indicates that they are not fully explained by arbitrage frictions.

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